
```
clc; clear; close all;
```

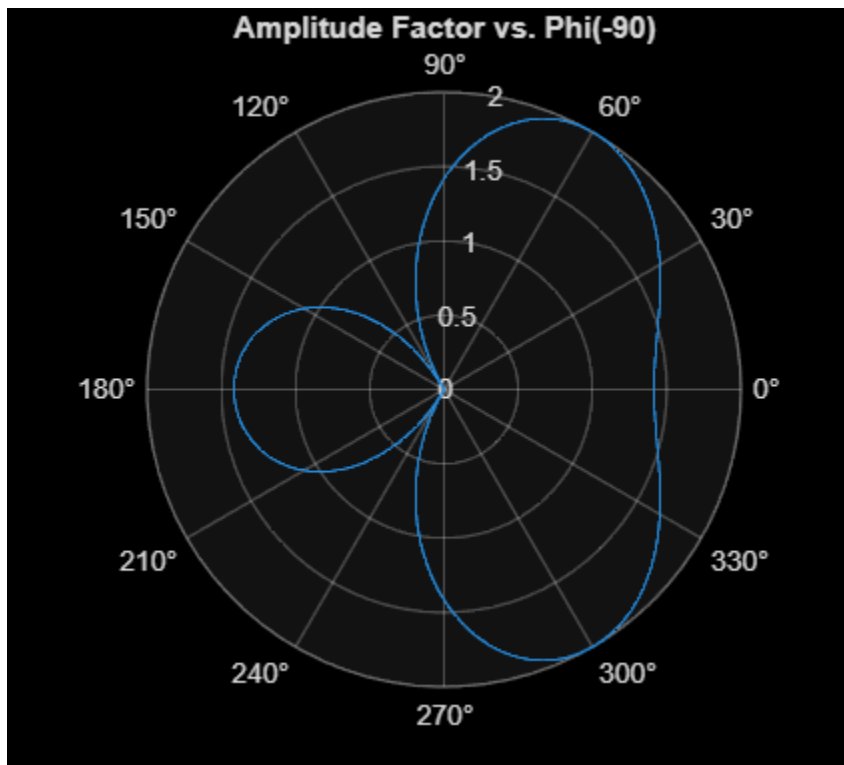
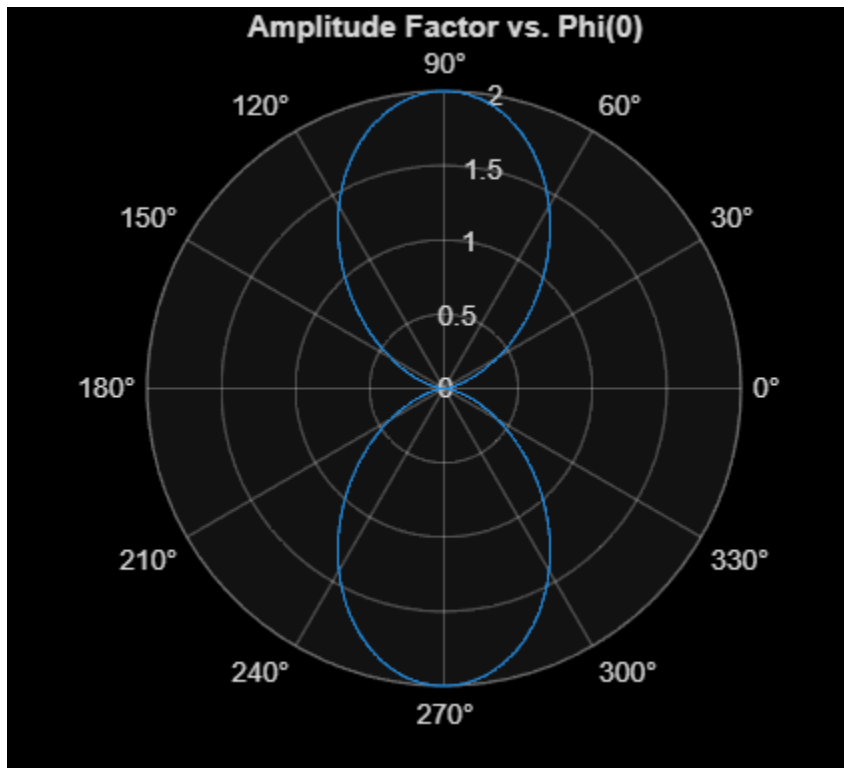
part a

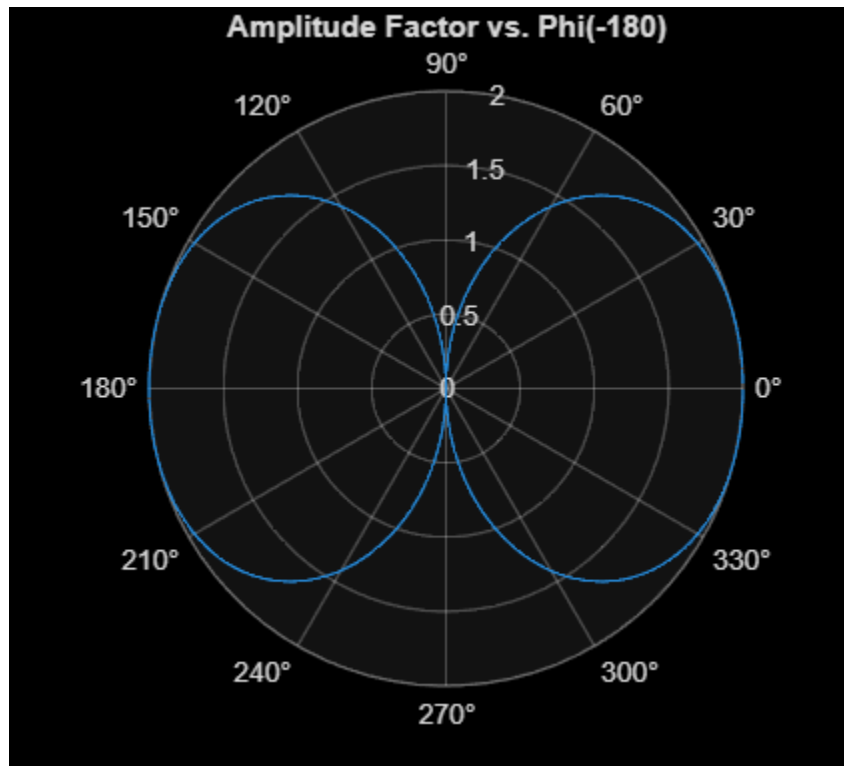
```
phi = linspace(0, 2*pi, 1000);
delta = 0;
AF = 1 + exp(1i*(pi*cos(phi)+delta)); %given equation
figure;
polarplot(phi, abs(AF)); %polarplot
title('Amplitude Factor vs. Phi(0)');
grid on;
phimax = phi(abs(AF) == max(abs(AF)));
fprintf('angle(s) of maximum |AF|[phi = 0] = %.2f degrees\n', rad2deg(phimax));
```

```
phi = linspace(0, 2*pi, 1000);
delta = -pi/2;
AF = 1 + exp(1i*(pi*cos(phi)+delta));
figure;
polarplot(phi, abs(AF));
title('Amplitude Factor vs. Phi(-90)');
grid on;
phimax = phi(abs(AF) == max(abs(AF)));
fprintf('angle(s) of maximum |AF|[phi = -90] = %.2f\n', rad2deg(phimax));
degrees\n', rad2deg(phimax));
```

```
phi = linspace(0, 2*pi, 1000);
delta = -pi;
AF = 1 + exp(1i*(pi*cos(phi)+delta));
figure;
polarplot(phi, abs(AF));
title('Amplitude Factor vs. Phi(-180)');
grid on;
phimax = phi(abs(AF) == max(abs(AF)));
fprintf('angle(s) of maximum |AF|[phi = -180] = %.2f\n', rad2deg(phimax));
degrees\n', rad2deg(phimax));
```

```
angle(s) of maximum |AF|[phi = 0] = 90.09 degrees
angle(s) of maximum |AF|[phi = 0] = 269.91 degrees
angle(s) of maximum |AF|[phi = -90] = 59.82 degrees
angle(s) of maximum |AF|[phi = -90] = 300.18 degrees
angle(s) of maximum |AF|[phi = -180] = 0.00 degrees
angle(s) of maximum |AF|[phi = -180] = 360.00 degrees
```





part b

```
phi = linspace(0, 2*pi, 10000);
phi_deg = rad2deg(phi);
delta = 0;
AF = 1 + exp(1i*(pi*cos(phi) + delta));
AF_mag = abs(AF);

AF_norm = AF_mag / max(AF_mag);
hp = 1/sqrt(2); % Half-power level
idx = find(abs(AF_norm - hp) < 0.01); % indices near half power
phi_hp = phi_deg(idx);
phi_hp = phi_hp(phi_hp > 0 & phi_hp < 180); %main lobe

phi1 = min(phi_hp);
phi2 = max(phi_hp);
HPBW = phi2 - phi1;
fprintf('Half-power angles: %.2f° and %.2f°\n', phi1, phi2);
fprintf('Numerical HPBW = %.2f°\n', HPBW);

Half-power angles: 59.44° and 120.58°
Numerical HPBW = 61.13°
```

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